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Experiment - 1 ⇒ Basic Networking Commands in Linux and Windows

Aim: To study and execute basic networking commands available in Linux and Windows operating systems for diagnosing and trouble shooting network issues.

Procedure:

* Open Command prompt or Terminal.
* Execute 'ipconfig' (Windows) or 'ifconfig' / 'ip a' (Linux) to view IP configuration.
* Use 'ping 8.8.8.8 to test connectivity to Google's DNS Server.
* Execute 'tracert google.com' (Windows) or 'traceroute google.com' (Linux) to view packet path.
* Record all output and analyze the result

Windows Commands

Commands	Discription	Sample O/P
ipconfig	- IP address	192.168.1.20
ipconfig /all	- MAC address, DHCP Server, DNS Server	windows ip configuration
Ping google.com	- Checks destination is reachable	Sent = 4 Received = 4 Lost = 0

tracert google.com	- displays every router between computer & dist	192.168.1.1 ISP Router Google Network.
nslookup google.com	- DNS server, IP address of website	Name: ... Address: ...
route print	- Network Routes, Gateway	Interface List
netstat	- Active TCP, Listening ports	Active connections TCP UDP
netstat -a	- All ports TCP connections UDP ports	active connections TCP UDP
getmac	- MAC address	PA TN
hostname	- computer name	DESKTOP-ABC123
arp -a	- IP to MAC address	Interface Address

Result:

Basic network commands were successfully executed on both Linux and Windows.

Linux Commands

```
[sj@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:12:f:dB:cc brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86069sec preferred_lft 86069sec
    inet6 fe80::a00:27ff:fe2f:d8cc/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[sj@localhost ~]$
```

```
[sj@localhost ~]$ hostname
localhost.localdomain
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ping google.com
PING google.com (142.251.43.142) 56(84) bytes of data.
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=1 ttl=118 time=6.27 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=2 ttl=118 time=7.93 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=3 ttl=118 time=11.1 ms
64 bytes from bkk03s01-in-f14.1e100.net (142.251.43.142): icmp_seq=4 ttl=118 time=5.11 ms
64 bytes from lcmaaa-aq-in-f14.1e100.net (142.251.43.142): icmp_seq=5 ttl=118 time=12.3 ms
```

```
[sj@localhost ~]$ nslookup google.com
Server:         8.8.8.8
Address:        8.8.8.8#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.205.110
Name:   google.com
Address: 2404:6800:4007:815::200e

[sj@localhost ~]$
```

```
[sj@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.locald:bootpc _gateway:bootps         ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp        0      0 localhost:323           0.0.0.0:*
udp6       0      0 [::]:mdns               [::]:*
udp6       0      0 localhost:323           [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
```

```
[sj@localhost ~]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:12:f:dB:cc brd ff:ff:ff:ff:ff:ff
[sj@localhost ~]$
```

```
[sj@localhost ~]$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default
qlen 1000
    link/ether 08:00:27:12:f:dB:cc brd ff:ff:ff:ff:ff:ff
[sj@localhost ~]$
[sj@localhost ~]$ ss -tuln
Netid      State      Recv-Q      Send-Q           Local Address:Port           Peer Address:Port
udp        ESTAB      0           0          10.0.2.15%enp0s3:68            10.0.2.2:67
[sj@localhost ~]$
```

```
[sj@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```